Themos cascode

$$g_{m2} V_{gs} = g_{m1} v_i \Rightarrow v_{gs2} = \frac{g_{m1} v_i}{g_{m2}}$$

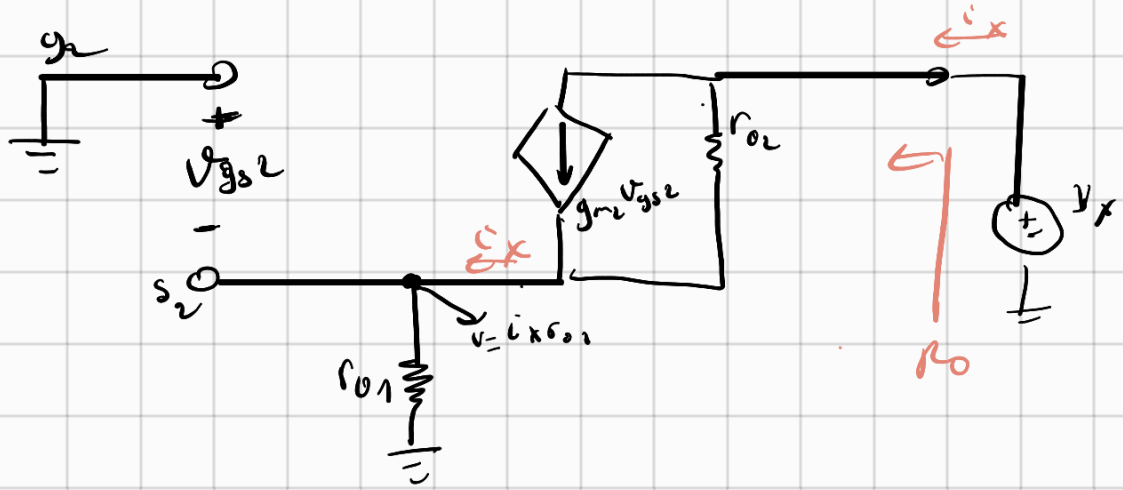
$$i_o = g_{m2} V_{gs2} + \frac{v_{gs2}}{r_{o2}}$$

$$i_o = g_{m2} \frac{g_{m1} v_i}{g_{m2}} + \frac{g_{m1} v_i}{g_m r_{o2}}$$

$$i_o = g_{m1} v_i \left(1 + 1/g_m r_{o2}\right)$$

$$\frac{i_o}{v_i} = g_{m1}$$

Now, replacing Q_2 with its hybrid π mode and applying a test voltage V_x to the output node results in the equivalent circuit shown.



$$R_o = \frac{v_x}{i_x} \qquad -v_{gs2} = i_x r_{o1}$$

$$v_x = \left(i_x - g_{m2} v_{gs2}\right) r_{o2} + i_x r_{o1}$$

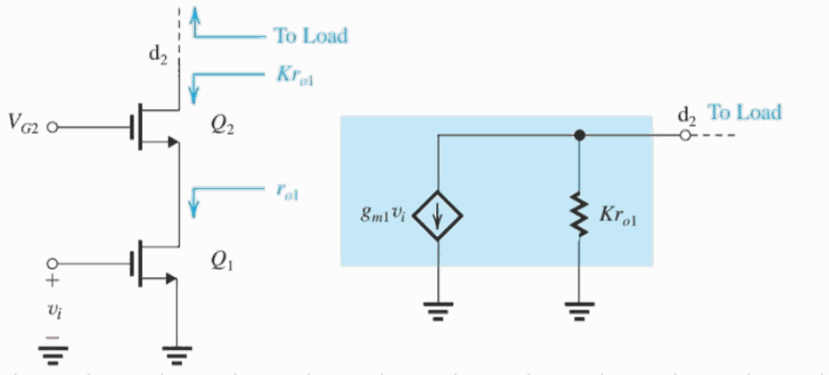
$$= i_x r_{o2} + g_{m2} i_x r_{o1} r_{o2} + i_x r_{o1}$$

$$\Rightarrow v_x = i_x \left(r_{o2} + r_{o1} + g_{m2} r_{o2} r_{o1}\right)$$

and $$R_o \equiv \frac{v_x}{i_x} = r_{o1} + r_{o2} + g_{m2} r_{o2} r_{o1}$$

$R_o \simeq (g_{m2} r_{o2}) r_{o1}$ because the last term will dominate

This expression has a simple and elegant interpretation
The CG Transistor Q_2 raises the output resistance of the amplifier by the factor $(g_{m2} r_{o2})$, which is its intrinsic gain.
At the same time, the CG transistor simply passes the current $(g_{m1} v_i)$ to the output node.
Thus the CG or cascode transistor very effectively realizes
The objectives we set for the current buffer with $k = A_{o2} = g_{m2} r_{o2}$



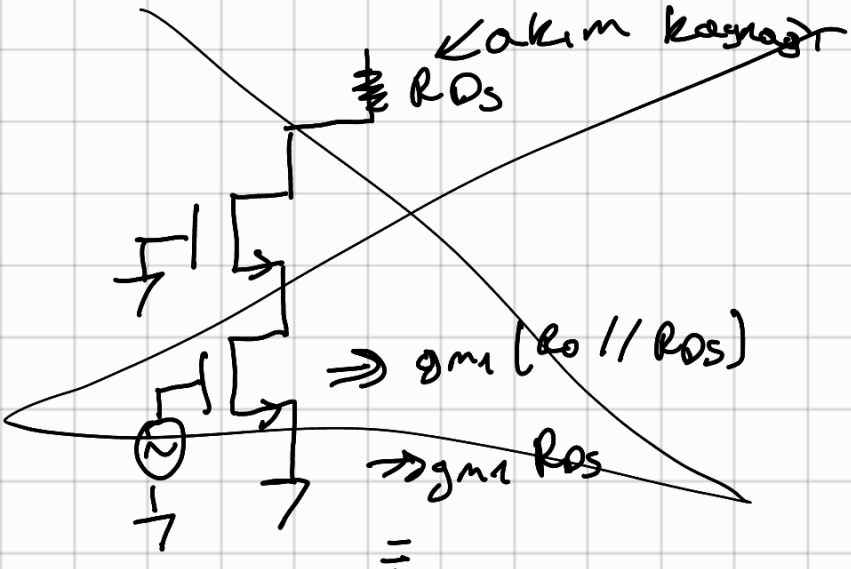
$$A_{vo} = \frac{v_o}{v_i} = \frac{-g_{m1} v_i R_o}{v_i} = -g_{m1} R_o$$

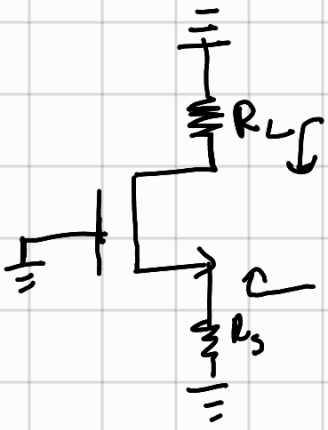
$$= -(g_{m1} r_{o1})(g_{m2} r_{o2})$$

For the case $g_{m1} = g_{m2} = g_m$ and $r_{o1} = r_{o2} = r_o$

$$A_{vo} = -(g_m r_o)^2 = A_o^2$$

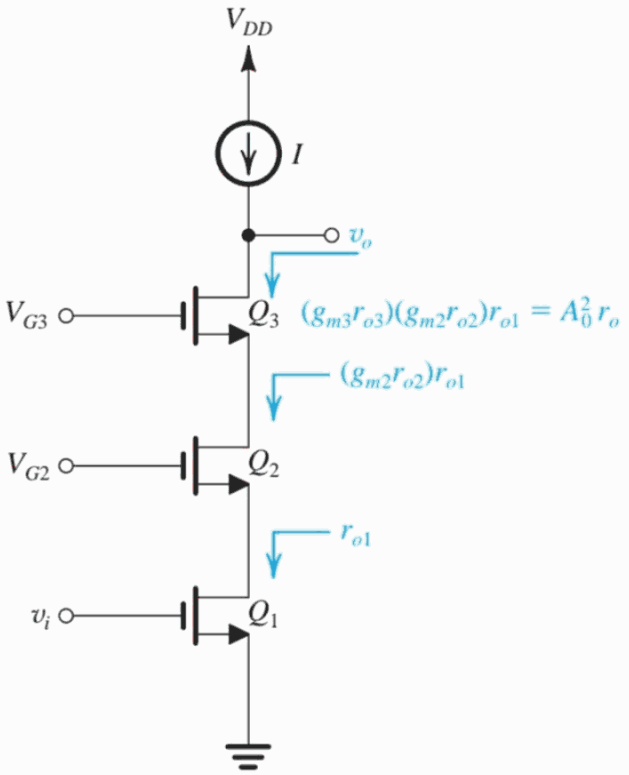
Thus cascoding results is increasing the gain magnitude from A_o to A_o^2





$$R_o = r_o + R_S + g_m r_o R_S$$
$$= r_o + (1 + g_m r_o) R_S$$

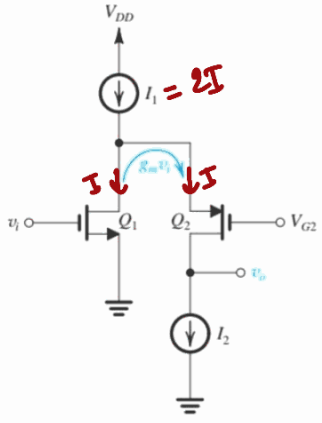
$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{(1 + g_m r_o)}$$



Supply gerilimi bizi limitliyor mosfetlerin üzerinden gerilim düşümünden dolayı

7.3.6 The Folded Cascode

To avoid the problem of stacking a large number of transistors across a low-voltage power supply, one can use a PMOS transistor for the cascode device, as shown in Fig. 7.16. Here, as before, the NMOS transistor Q_1 is operating in the CS configuration, but the CG stage is implemented using the PMOS transistor Q_2. An additional current source I_2 is needed to bias Q_2 and provide it with its active load. Note that Q_1 is now operating at a bias current of $(I_1 - I_2)$. Finally, a dc voltage V_{G2} is needed to provide an appropriate dc level for the gate of the cascode transistor Q_2. Its value has to be selected so that Q_2 and Q_1 operate in the saturation region.



2kat gücü harcar

Figure 7.16 The folded cascode.

The BJT Cascode

7.3.7 The BJT Cascode

Figure 7.17(a) shows the BJT cascode amplifier with an ideal current-source load. Voltage V_{B2} is a dc bias voltage for the CB cascode transistor Q_2. The circuit is very similar to the MOS cascode, and the small-signal analysis will follow in a parallel fashion. Our objective then is to determine the parameters G_m and R_o of the equivalent circuit of Fig. 7.17(b), which we shall use to represent the output of the cascode amplifier formed by Q_1 and Q_2.

As in the case of the MOS cascode, G_m is the short-circuit transconductance and can be determined from the circuit in Fig. 7.17(c). Here we show the cascode amplifier prepared for small-signal analysis with the output short-circuited to ground. The transconductance G_m can be determined as

$$G_m = \frac{i_o}{v_i}$$

Replacing Q_1 and Q_2 with their hybrid-π equivalent-circuit models gives rise to the circuit in Fig. 7.17(d). Analysis of this circuit is straightforward and proceeds as follows: The voltage at the node (c_1, e_2) is seen to be $-v_{\pi 2}$. Thus we can write a node equation for (c_1, e_2) as

$$g_{m2}v_{\pi 2} + \frac{v_{\pi 2}}{r_{o1}} + \frac{v_{\pi 2}}{r_{o2}} + \frac{v_{\pi 2}}{r_{\pi 2}} = g_{m1}v_i$$

[3]The circuit itself can be thought of as having been folded. In this same vein, the regular cascode is sometimes referred to as a **telescopic cascode** because the stacking of transistors resembles the extension of a telescope.

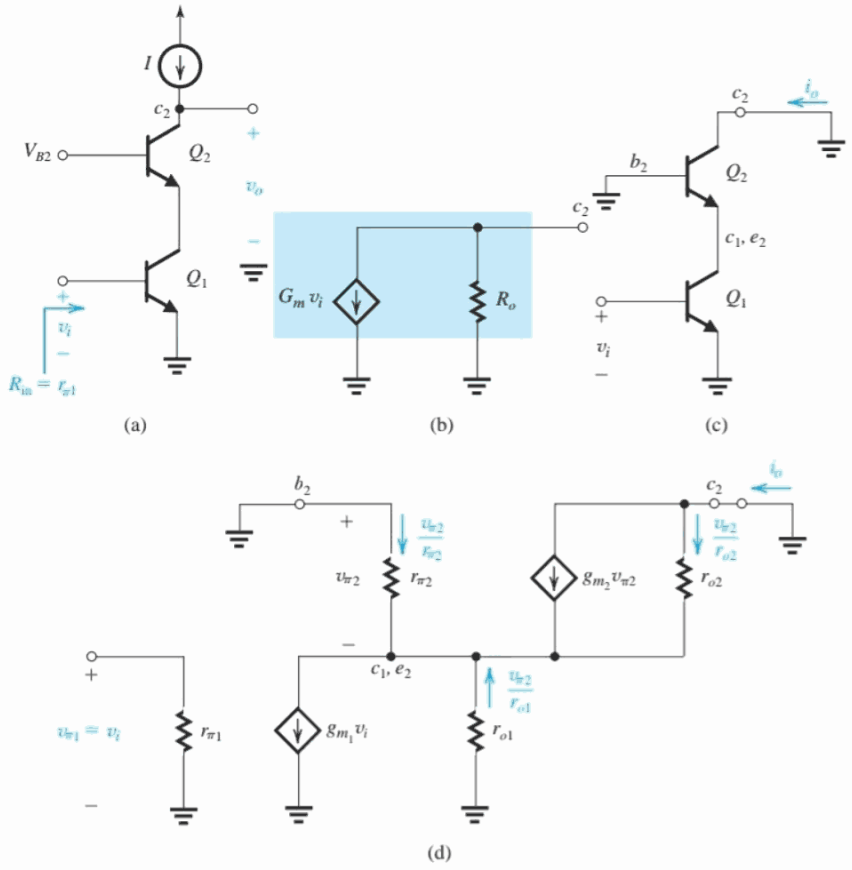


Figure 7.17 **(a)** A BJT cascode amplifier with an ideal current-source load; **(b)** small-signal equivalent-circuit representation of the output of the cascode amplifier; **(c)** the cascode amplifier with the output short-circuited to ground, and **(d)** equivalent circuit representation of **(c)**.

Since $g_{m2} \gg (1/r_{\pi 2})$, $1/r_{o1}$ and $1/r_{o2}$, we can neglect all the terms beyond the first on the left-hand side to obtain

$$g_{m2} v_{\pi 2} \simeq g_{m1} v_i \tag{7.41}$$

Next, we write a node equation at c_2,

$$i_o = g_{m2} v_{\pi 2} + \frac{v_{\pi 2}}{r_{o2}}$$

and again neglect the second term on the right-hand side to obtain

$$i_o \simeq g_{m2} v_{\pi 2}$$

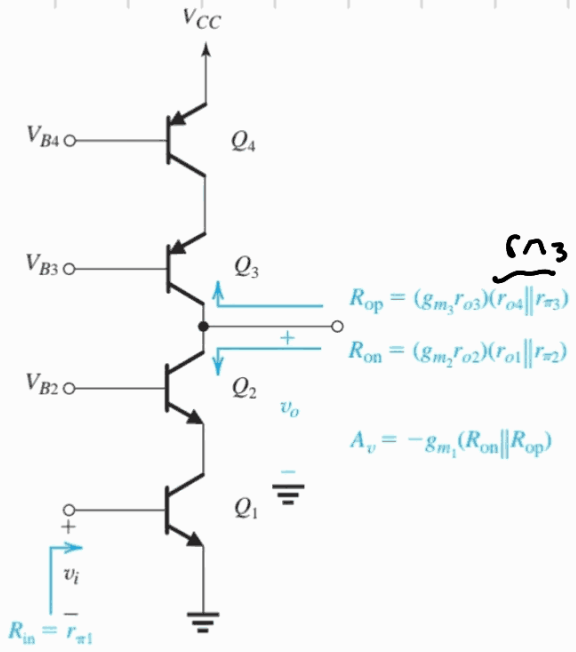
Using Eq. (7.41) results in

$$i_o = g_{m1} v_i$$

$r_\pi = (\beta+1) \| r_e$

$g_m = I_{CQ}/V_T$

$r_\pi = \left(\frac{\beta+1}{g_m}\right)$



because $r_{o4} \gg r_{\pi 3}$

$r_{\pi 3}$

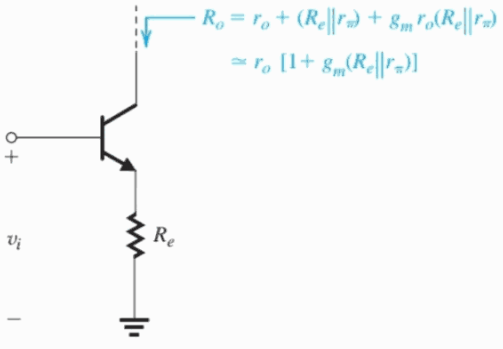
$r_{\pi 3} = g_{m3} r_{o3} \frac{\beta+1}{g_{m3}} \simeq \beta r_{o3}$

$r_\pi = (\beta + 1) r_e$

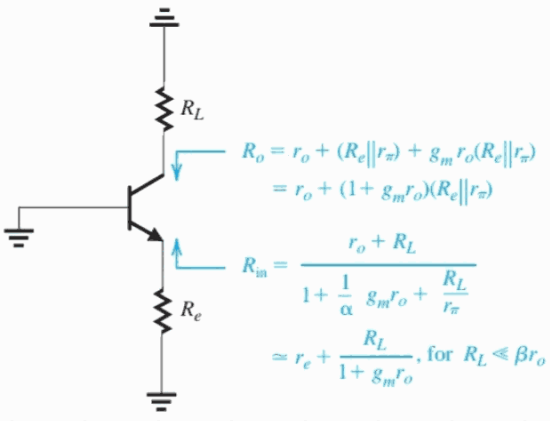
$r_e = \frac{V_t}{I_E}$

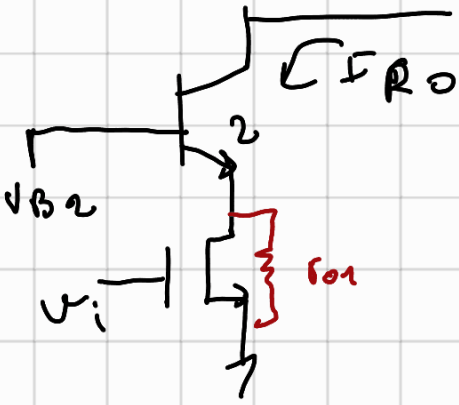
$g_m = \frac{I_C}{V_t}$

$r_e = 1/g_m$



(a)

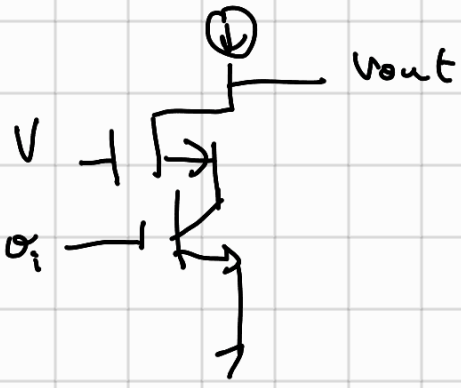


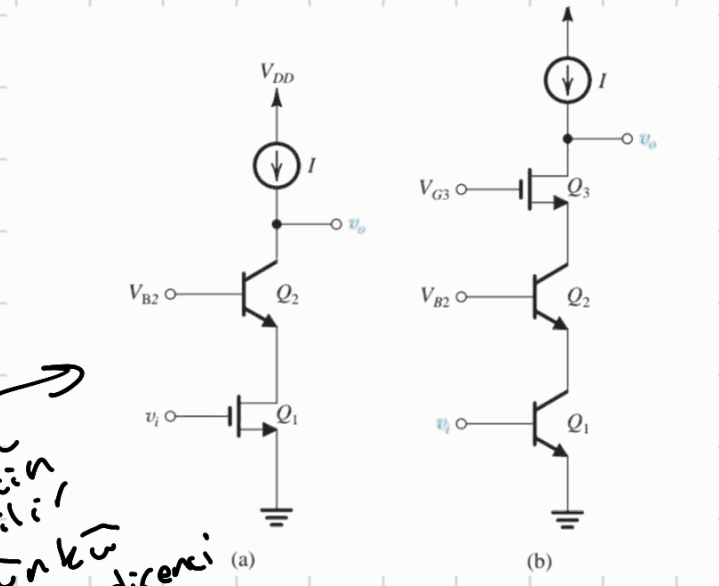


$= r_{o2} + (1 + g_m r_o)(R_E \,//\, r_{o1})$

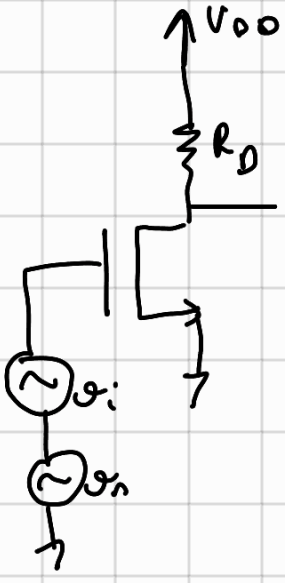
$= r_{o2} + (1 + g_m r_{o2})(r_{o1} \,//\, r_{o1})$

BiCMOS cascodes



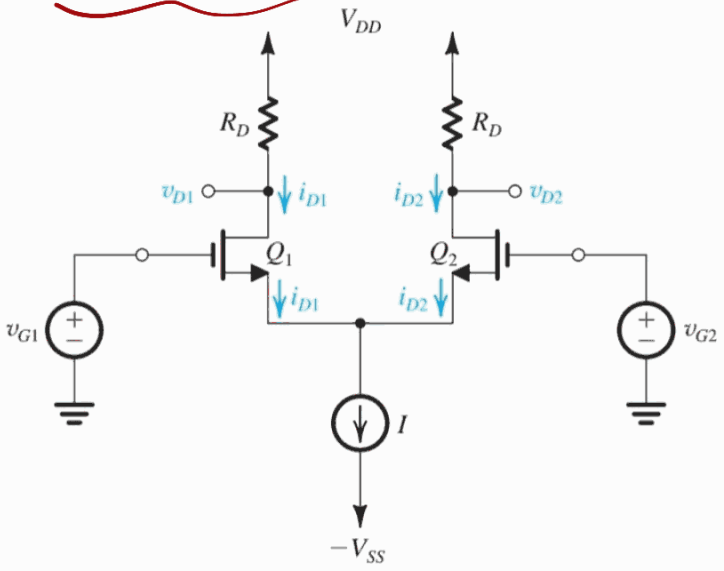


Du tercih edilir çünkü giriş direnci sonsuz



$V_{out} = -g_m (R_D \,//\, r_o)(v_i + v_n)$

Differential amplifier



$$v_{G1} = \frac{v_i}{2} + v_n$$

$$v_{G2} = -\frac{v_i}{2} + v_n$$

$$V_{diff\,out} = A_V (V_{G1} - V_{G2})$$

$$\frac{I}{2} = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

$$V_{OV} = V_{GS} - V_t$$

$$\frac{I}{2} = \frac{1}{2} k'_n \frac{W}{L} V_{OV}^2$$

$$V_{OV} = \sqrt{\frac{I}{k'_n (W/L)}}$$

$$V_{D1} = V_{D2} = V_{DD} - \frac{I}{2} R_D$$

$$\max(V_{CM}) = V_t + V_{DD} - \frac{I}{2} R_D$$

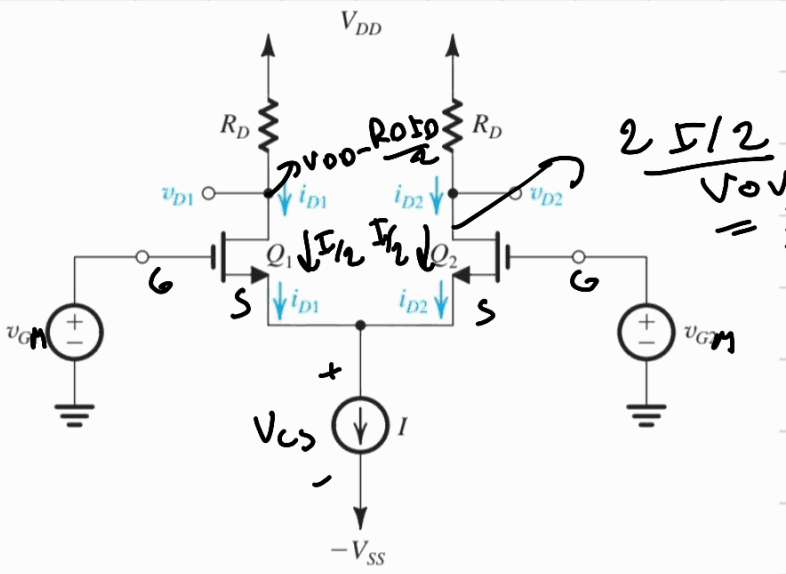
$$\min(V_{CM}) = -V_{SS} + V_{GS} + V_t + V_{OV}$$

$$V_{GS} > V_t$$

$$V_{DS} > V_{GS} - V_{tn}$$

$$V_{CM_{max}} = V_D + V_{tn}$$

$$\| V_{GS}$$



$V_{DD} - R_D \frac{I}{2}$

$\frac{2 I/2}{V_{OV}} = \frac{I}{V_{OV}}$

$v_{D2} - v_{D1} = 0$ always

$V_{GS} \geq V_{tn}$

$V_{CM} - (-(V_{CS} + V_{SS})) \geq V_{tn}$

$V_{CM} \geq -V_{SS} + V_{CS} + V_{tn}$

$V_{DS} \geq V_{GS} - V_{tn}$

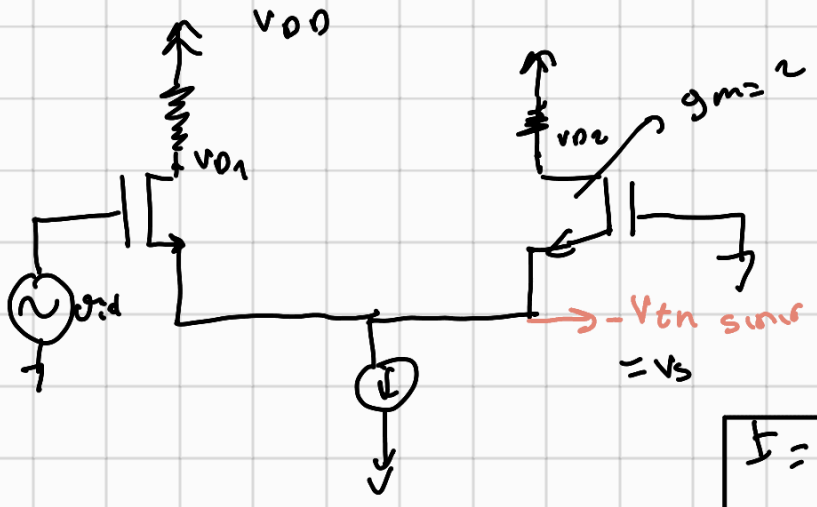
$V_D \geq V_G - V_{tn}$

$V_D - V_{tn} \geq V_{CM}$

$\max(V_{CM}) = V_D - V_{tn}$

$= V_{DD} - \frac{I}{2} R_D - V_{tn}$

$-V_{SS} + V_{CS} + V_{tn} < V_{CM} < V_{DD} - \frac{I}{2} R_D - V_{tn}$



$V_{GS_2} = V_{tn}$ → limit condition

$$I = \frac{1}{2} k'_n \frac{W}{L} (V_{GS_1} - V_{tn})^2$$

$$v_{GS_1} = \sqrt{2}\, V_{ov} + V_{th}$$

$$v_{id_{max}} = \sqrt{2}\, V_{ov}$$

$$v_{id_{min}} = -\sqrt{2}\, V_{ov}$$

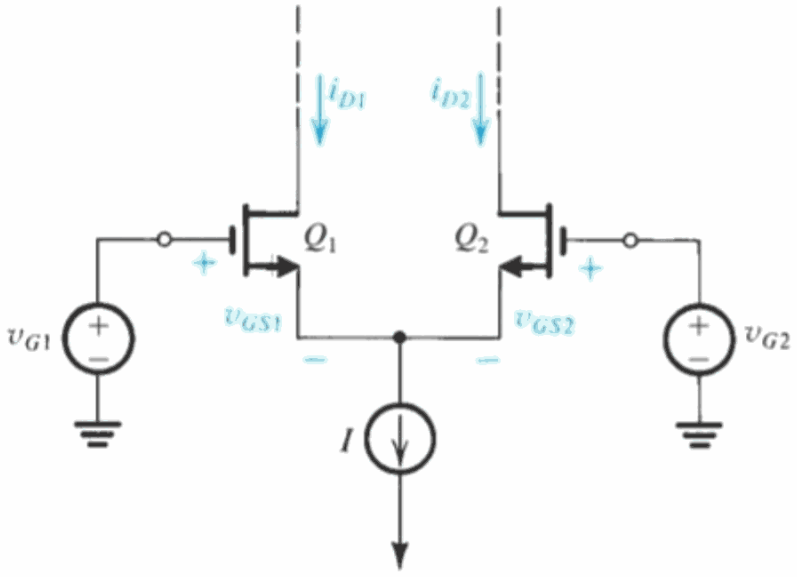


Figure 8.5 The MOSFET differential pair for the purpose of deriving the transfer characteristics, i_{D1} and i_{D2} versus $v_{id} = v_{G1} - v_{G2}$.

these connections in Fig. 8.5; we simply assume that the circuit maintains Q_1 and Q_2 in the saturation region of operation at all times.

To begin with, we express the drain currents of Q_1 and Q_2 as

$$i_{D1} = \frac{1}{2} k_n' \frac{W}{L} (v_{GS1} - V_t)^2 \tag{8.11}$$

$$i_{D2} = \frac{1}{2} k_n' \frac{W}{L} (v_{GS2} - V_t)^2 \tag{8.12}$$

Taking the square roots of both sides of each of Eqs. (8.11) and (8.12), we obtain

$$\sqrt{i_{D1}} = \sqrt{\frac{1}{2} k_n' \frac{W}{L}} (v_{GS1} - V_t) \tag{8.13}$$

$$\sqrt{i_{D2}} = \sqrt{\frac{1}{2} k_n' \frac{W}{L}} (v_{GS2} - V_t) \tag{8.14}$$

Subtracting Eq. (8.14) from Eq. (8.13) and substituting

$$v_{GS1} - v_{GS2} = v_{G1} - v_{G2} = v_{id} \tag{8.15}$$

results in

$$\sqrt{i_{D1}} - \sqrt{i_{D2}} = \sqrt{\frac{1}{2} k_n' \frac{W}{L}} v_{id} \tag{8.16}$$

The constant-current bias imposes the constraint

$$i_{D1} + i_{D2} = I \tag{8.17}$$

Equations (8.16) and (8.17) are two equations in the two unknowns i_{D1} and i_{D2} and can be solved as follows: Squaring both sides of Eq. (8.16) and substituting for $i_{D1} + i_{D2} = I$ gives

$$2\sqrt{i_{D1} i_{D2}} = I - \frac{1}{2} k_n' \frac{W}{L} v_{id}^2$$

Substituting for i_{D2} from Eq. (8.17) as $i_{D2} = I - i_{D1}$ and squaring both sides of the resulting equation provides a quadratic equation in i_{D1} that can be solved to yield

$$i_{D1} = \frac{I}{2} \pm \sqrt{k_n' \frac{W}{L} I} \left(\frac{v_{id}}{2}\right) \sqrt{1 - \frac{(v_{id}/2)^2}{I/k_n' \frac{W}{L}}}$$

Now, since the increment in i_{D1} above the bias value of $I/2$ must have the same polarity as v_{id}, only the root with the "+" sign in the second term is physically meaningful; thus,

$$i_{D1} = \frac{I}{2} + \sqrt{k_n' \frac{W}{L} I} \left(\frac{v_{id}}{2}\right) \sqrt{1 - \frac{(v_{id}/2)^2}{I/k_n' \frac{W}{L}}} \tag{8.18}$$

The corresponding value of i_{D2} is found from $i_{D2} = I - i_{D1}$ as

$$i_{D2} = \frac{I}{2} - \sqrt{k_n' \frac{W}{L} I} \left(\frac{v_{id}}{2}\right) \sqrt{1 - \frac{(v_{id}/2)^2}{I/k_n' \frac{W}{L}}} \tag{8.19}$$

At the bias (quiescent) point, $v_{id} = 0$, leading to

$$i_{D1} = i_{D2} = \frac{I}{2} \tag{8.20}$$

Correspondingly,

$$v_{GS1} = v_{GS2} = V_{GS} \tag{8.21}$$

where

$$\frac{I}{2} = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 = \frac{1}{2} k_n' \frac{W}{L} V_{OV}^2 \tag{8.22}$$

This relationship enables us to replace $k_n'(W/L)$ in Eqs. (8.18) and (8.19) with I/V_{OV}^2 to express i_{D1} and i_{D2} in the alternative form

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right) \sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2} \quad v_{id} \ll V_{OV} \tag{8.23}$$

$$i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right) \sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2} \tag{8.24}$$

These two equations describe the effect of applying a differential input signal v_{id} on the currents i_{D1} and i_{D2}. They can be used to obtain the normalized plots, i_{D1}/I and i_{D2}/I versus

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

$$i_2 = I/2 - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

$$v_{O1} = V_{DD} - R_D i_{D1}$$

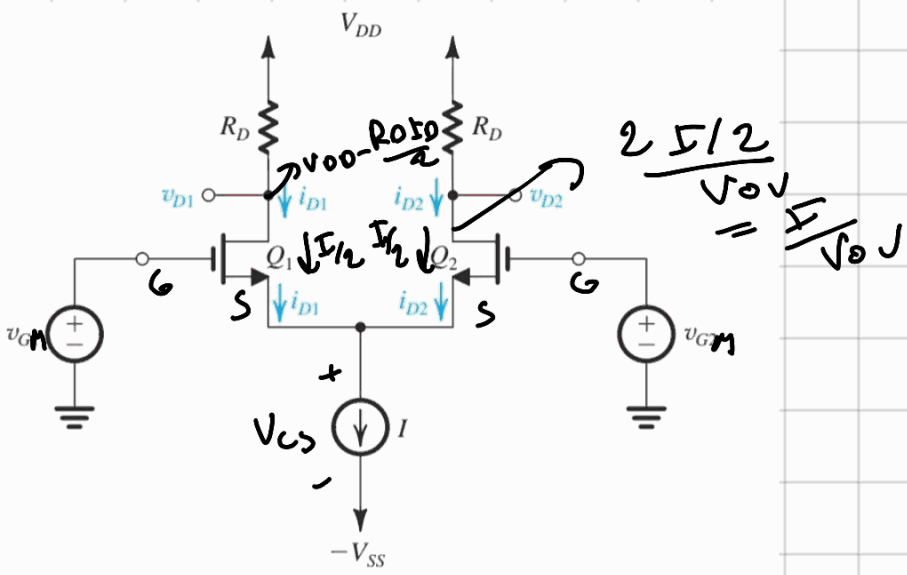
$$v_{O2} = V_{DD} - R_D i_{D2}$$

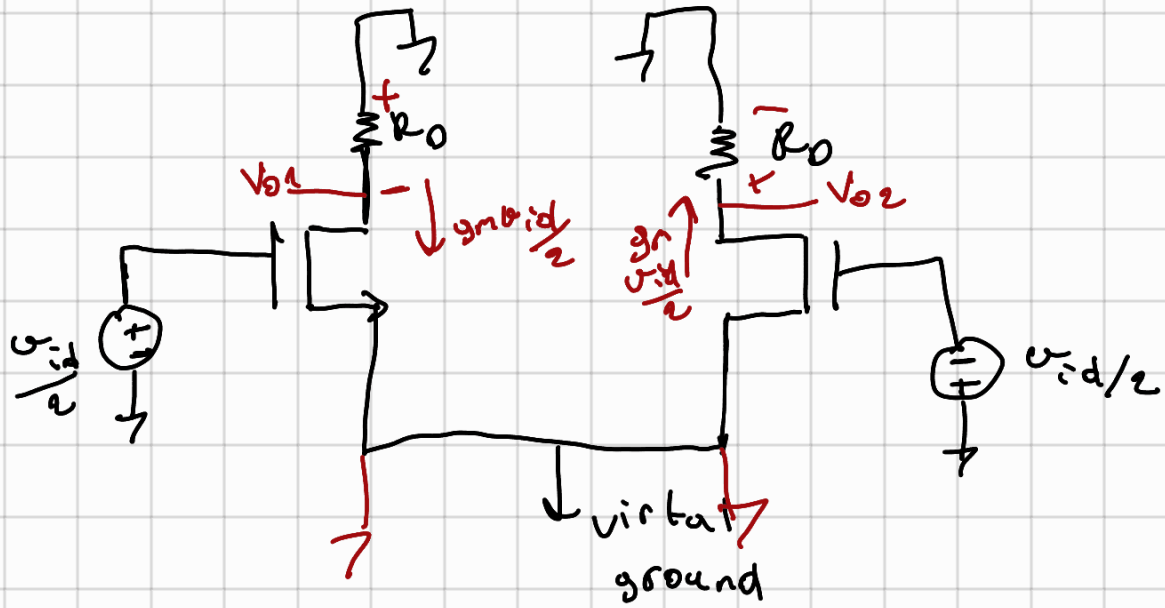
$$v_{O2} - v_{O1} = R_D i_{D1} - R_D i_2 = R_D (i_{D1} - i_{D2})$$

$$v_{out} = v_{O2} - v_{O1} = R_D \left[\frac{I}{V_{OV}} \cdot v_{id}\right]$$

$$\frac{v_{out}}{v_{id}} = R_D \frac{I}{V_{OV}} \quad v_{id}$$

$$\frac{v_{out}}{v_{id}} = R_D \frac{I}{V_{OV}} = g_m R_D$$





ayrı ayrı çözümlemi iş

